

AP EAMCET AGRICULTURE AND MEDICINE

Year – 2017

No. of Question: 160

1. Study the following lists

List -I	List-II
A) Convolvulaceae	i) Dicotyledons
B) Rosales	ii) Arthropoda
C) Mangifera	iii) Anacardiaceae
D) insecta	iv) Polemoniales
	v) Monocotyledons

The collect answer is

Options:

A B C D

a) iv i v ii

b) iv i iii ii

c) iv i v iii

d) iv i ii iii

2. Choose the collect statement from the following

- I. Rhodophyceae cell wall is made of cellulose, pectin and algin
- II. Marchantia disperse spores by pseudoelaters
- III. Diyopteris shows circinate veneration
- IV. Primary endospeimic nucleus is triploid

Options:

- a) I, II
- b) II, III
- c) I, IV
- d) III, IV

3. Choose the following characteristics found in Noctiluca:

- A. Bioluminescence
- B. Prokaryote
- C. Proteinaceous cell wall
- D. Cell wall with cellulose plates
- E. Chromosome without histones
- F. Eukaryote
- G. Longitudinal binary fission

options

- a) A, B, D, E, G
- b) A, C, E, F
- c) D, E, E, G
- d) A, D, E, F

4. Find the correct statement

- A) Moss plant is found in cool damp and shady place embryophytic. archegoniate and vascular in nature.
- B) Majority of red algae are marine and found in deep waters as well as shallow areas
- C) Algae constitute the half of carbon dioxide fixation on earth and also form the basis for food cycle

D) In Gymnosperms the transfer of pollen to the stigma of the Gynoecium in the megasporophyll is called direct pollination

Options:

- a) A, B
- b) B, C
- c) C, D
- d) A, C

5. Numerically identify the given group of plants based on fruits formed from the entire inflorescence and type indehiscent fruits types respectively

Custard Apple	Tridax	Pineapple	Cashew Nut	Rice
Jack Fruit	Beans	Acacia		Datura

Options:

- a) 2,3
- b) 3,6
- c) 3,5
- d) 2,4

6.

- A. Flowers are brought to the same height though they born at different nodes
- B. Sessile unisexual flowers arranged in acropetal succession
- C. Presence of involucre Based on above inflorescence

characters A B and C plants are respectively identified as

Options:

- a) Carrot, Cauliflower, Colacasia
- b) Carrot, Colacasia, Cauliflower
- c) Cauliflower, Carrot, Colacasic
- d) Cauliflower, Colacasia, Carrot

7. Study the following

- A. Apomicts do not have the evolutionary advantage of sexual reproduction
- B. Apomictically produced off springs are genetically identical to parent plants
- C. Apomixis is a form of sexual reproduction
- D. Apomixis is an assured reproduction in some species of Solanaceae

Find correct statements.

Options:

- a) A, D
- b) A, C
- c) B, D
- d) A, B

8. Find the wrong statements among the following:

- A. Seeds of groundnut and bean are nonendospermic
- B. Stomium is useful for the development of pollen tubes in

suitable medium

C. Tagetacula helps in pollination of Amorphophallus

D. Seeds have to be dehydrated to raise the next season crop plants

Options :

a) A,B,C

b) B,C

c) A,C,D

d) A,D

9. Study the lists

List -1	List -II	List - III
A) Microsporogenous tissue	i) Diploid	I) Pollengrain
B) Endothecium	ii) Triple fusion	II) Microspore mother cell
C) Generative cell	iii) Pollen chamber	III) Triploid
D) Second male gamete	iv) Haploid	IV) Dehiscence

Study the lists

	A	B	C	D
a)	ii, III	iii, IV	i, II	iv, I
b)	iii, II	ii, IV	iv, I	i, III
c)	iii, II	iv, III	ii, I	I, IV
d)	iii, II	i, IV	iv, I	ii, III

10. Match The following lists

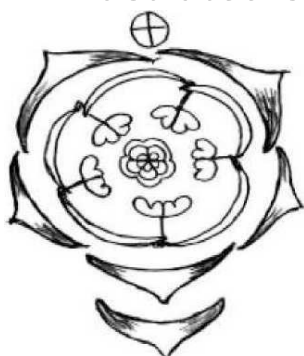
List -1	List - II
A) Monocarpellary ovary	I) Papaver
B) Apocarpous ovary	II) Dolichos
C) Syncarpous ovary	III) Allium
D) Tri carpellary ovary	IV) Michelia
	V) Abutilon

The correct answer is

Options:

- A B C D
- a) II I IV III
- A B C D
- b) II IV I III
- A B C D
- c) II V I III
- A B C D
- d) II III IV I

11. Find the collect description of the flower in general and 1st and 3rd whorl of flower based on the diagram illustrated below.



Options:

- a) Bracteate, ebracteolate, pentamerous, bisexual, Gamosepalous. twisted aestivation, stamens 5, Epipetalous
- b) Ebracteate, ebracteolate pentamerous, bisexual, polysepalous, imbricate aestivation, stamens 5, epipliyllous
- c) Ebracteate, ebracteolate, pentamerous, bisexual, gamosepalous, valvate aestivation, stamens 5, monoadelphous
- d) Bracteate, ebracteolate, pemanierous, bisexual, polysepalous, imbricate aestivation, stamens 5, epipetalous

12. If a megaspore mother cell of Pisimi Sativam undergoes meiosis. the ratio of no. of chromosomes and chromatids in different phases of cell division is

- Chromosomes in Metaphase I
- Chromosomes at one pole of Anaphase I
- Chromatids at one pole of Metaphase II
- Chromosomes in Anaphase n

Options:

- a) 14:7:7:28
- b) 14:14:7:14
- c) 14:7:14:28
- d) 14:14:7:28

13. Study the following table

A) Peroxisomes	Glyoxysomes	Microbodies
B) Flagella	Peripheral Micro tubules	9 pairs
C) Cytoskeleton	Eukaryotic cell	Cell motility
D) Chloroplasts	Carotenoid pigments	Water soluble

The correct combination are

Options:

- a) A, B, D
- b) A, B, C
- c) B, C, D
- d) A, C, D

14. Match the given bonding groups and their respective bond.

List - I	List - II
[Bonding groups]	[Bonds]
A) 2C-atoms of adjacent monosaccharides	I) Ionic bond
B) Carboxyl and amino group of two amino acids	II) Hydrogen bond
C) Two nucleotides	III) Peptide bond
D) Phosphate and hydroxyl group of sugar in DNA	IV) Ester bond
	V) Glycosidic bond

The correct answer is

- | | A | B | C | D |
|----|---|-----|-----|-----|
| a) | I | II | III | IV |
| b) | I | III | II | IV |
| c) | I | IV | II | III |
| d) | I | II | IV | III |

15. Match the following

List -1	List - H	List-HI
A) Metabolic pathways	I) Abundant protein	i) Biologically active proteins
B) Metabolism	II) Criss-Cross and dynamic	ii) Animals/ plants
C) Collagen / Rubisco	III) Hollow in nature	iii) Mechanism for energy production
D) Tertiary structure of protein	IV) Cannot reach equilibrium	iv) Catalysed reaction

The correct answer is

- | | A | B | C | D |
|----|-------|---------|--------|-------|
| a) | I,iv | II,iii | III,ii | IV,i |
| b) | II,iv | III,iii | I,ii | IV,i |
| c) | I,ii | IV,iii | I,ii | III,i |
| d) | II,iv | IV,iii | I,ii | III,i |

16. Match the given aspects shown by a DNA molecule having length A and 40% cytosine.

List -1	List - II
---------	-----------

A) No. of turns	I) 128
B) No. of H ₂ bonds	II) 318
C) No. of pyrimidines	III) 368
D) No. of phosphodiester bonds	IV) 16
	V) 32

The correct answer is

- | | A | B | C | D |
|----|----|-----|----|-----|
| a) | I | III | IV | II |
| b) | IV | II | I | III |
| c) | IV | III | I | II |
| d) | II | III | IV | I |

17. Choose wrong pair from the following

A) Theodore Schwann	Animal cell	plasma membrane
B) Vesicles, lamellae, tubules	prokaryote	contain pigments
C) Phosphate granule	Eukaryote	lysosomes
D) Nucleosome	Eukaryote	DNA packaging in nucleus

Options:

- a) A,B
- b) B,C
- c) A,C
- d) A,D

18. Identify the correct series of tissues showing the following
- Highly thickened spherical dead cells with lumen
 - Spherical cells with chloroplasts without intercellular spaces
 - Longitudinal cells, without nucleus, large vacuole and peripheral cytoplasm
 - Elongated cells with thick lignified walls and have variable thickenings with inner layer.

Options :

- Sclereids. Parenchyma. Fibres. Albuminous Cells
- Sieve cells. Collenchyma. Fibres. Sievetubes
- Sclereids. Collenchyma. Sieve tube. Tracheids
- Sieve cells, parenchyma. Sieve tube. Sclereids

19. Identify the correct statement from the following.

- Hypodermis is sclerenchymatous in monocot stem where as collenchymatous in dicot stem.
- Vascular bundles are scattered in monocot stem where as arranged in a ring in dicot stem.
- Vascular bundle is surrounded by collenchymatous sheath in monocot stem but by sclerenchymatous sheath in dicot stem.
- Phloem parenchyma is absent in both monocot and dicot stem.

Options :

- I, II, III, IV

- b) I,II,III
- c) I,III
- d) I,II

20. Identify wrong statements regarding hydrophytes and xerophytes

	Hydrophytes	Xerophytes
A Root caps	Present	Well developed
B. Epidermis	Thin and do not contain chloroplasts	Thick and have chloroplasts silica crystals
C. Leaves	Well developed xylem	Single layed epidermis
D. Stomata	Upper surface	Lower surface

Options :

- a) A,C
- b) B,C
- c) A,B
- d) C,D

21. Two potato pieces A and B weighing 5 grams each were soaked in 0.1M and 0.5M sucrose solutions respectively. After an hour 'A' weighed 5.5 grams and B 4.5 grams. Reasons for the change in weights of A and B.

- A. Chemical potential of water is high in A than B
- B. Osmotic potential is high in B than A
- C. Turgor pressure is liigh in A than B
- D. Pressure potential in 'A' is 'O'

Choose the correct answer

Options :

- a) A,B,C,D
- b) A,B,C
- c) B,C,D
- d) B,D

22. Find out number of protons. electrons, and ATP molecules required and no. of ADP molecules released for the formation of 136 Ammonia molecules by symbiotic N_2 fixation, based on the equation given below.



Options:

- a) 272. 544. 1088. 544
- b) 544. 544. 1088. 1088
- c) 544. 272. 544. 1088
- d) 1088. 1088. 544. 544

23. Find the incorrect statements

- A. Malonate inhibits the succinic dehydrogenase action as a competitive inhibitor
- B. Enzyme that catalyses peptide bonds. C -halide bonds and P-N bonds are called lyases
- C. Thermophilic organism show thermal stability
- D. The difference between average energy content and that of the transition state is called transition energy

The correct answer is

Options :

- a) B,C
- b) A,D
- c) A,C
- d) B,D

24. Correct sequence of electron carriers in modern photosynthetic electron transport

Options:

- a) $H_2O \rightarrow Chl_{680} \rightarrow Pheo \rightarrow PQ \rightarrow Cytb_6 f \rightarrow P_{700} \rightarrow FD \rightarrow FNR$
- b) $H_2O \rightarrow P_{680} \rightarrow Pheo \rightarrow Cytb_6 \rightarrow PQ \rightarrow Cyt f \rightarrow P_{700} \rightarrow FD \rightarrow FNR$
- c) $H_2O \rightarrow P_{680} \rightarrow Pheo \rightarrow Cytb_6 \rightarrow Cyt f \rightarrow PQ \rightarrow P_{700} \rightarrow FD \rightarrow FNR$
- d) $H_2O \rightarrow P_{700} \rightarrow FND \rightarrow Pheo \rightarrow Cytb_6 \rightarrow Cyt f \rightarrow P_{680} \rightarrow H_2O$

25. Identify the correct match

List -1	List - II	List - III
A) Light dependent reaction	i) Sugars are not synthesised	I) CO ₂ , absorption
B) Photorespiration	ii) KOH	II) CO ₂ , reduction
C) Light independent reaction	iii) Photolysis of water	III) O ₂ , competes with CO ₂ ,
D) Half-leaf experiment	iv) Synthesis of sugars	IV) Reduction of NADP

The correct answer is

- | | A | B | C | D |
|----|--------|--------|--------|-------|
| a) | iii,IV | i,III | iv,II | ii,I |
| b) | iii,I | I,II | iv,III | ii,I |
| c) | iii,I | i,II | iv,III | ii,IV |
| d) | ii,IV | iv,III | i,II | iii,I |

26. Study the given table showing Kreb's cycle reactions and choose correct pairs

Reactant	No. of C in end product	Type of reaction
A) Oxalosuccinic acid	5	Decarboxylation
B) Succinic acid	4	Dehydrogenation
C) Oxaloceric acid	5	Condensation
D) Isocitric acid	6	Oxidation

Options:

- a) A,B
- b) B,C,D
- c) A,B,D
- d) A,C,D

27. Arrange the given list of processes that constitute the development of cells in higher plants in correct sequence.

- A) Senescence
- B) Elongation
- C) Maturation
- D) Differentiation

Arrange the above in a correct sequence

Options :

- a) B,C,D,A
- b) A,C,D,B
- c) B,D,C,A
- d) A.B.D,C

28. Choose the wrong statement

Options :

- a) The meristematic cells are rich with protoplasm.
Increased vacuolation makes the cell to enlarge.
- b) In the initial stages of development zygote shows a region of shoot shows arithmetic growth where apical region of shoot shows geometrical growth
- c) Sigmoid curve is observed in plants growing in natural conditions.
- d) The relative growth rate is also referred as efficiency index IANA

29.

- A. A replica of plasmid is donated to recipient by donor during Bacterial conjugation.
- B. Time taken for transfer of donor DNA to recipient bacterium is called generation time
- C. True sexual reproduction is absent in bacteria
- D. DNA components of bacteria can detect biologically active toxic pollutants

Identify incorrect statements from the above

Options :

- a) A,B
- b) A,B,C
- c) A,B,C,D
- d) A,B,D

30. Measles virus and bacteriophages possess following characters

Options :

- a) Glycoprotein without spikes and Helical symmetry
- b) Glycocalyx and Helical Symmetry
- c) Spikes and Spherical
- d) Glycoprotein with spikes and complicated structure

31. If a round seeded (BB) Pea with large starch grain was crossed with wrinkled seeded (bb) pea with small starch grain. Find the true reasons for F₁ off springs showing round seed (Bb) with intermediate starch grain

Following true inferences are drawn from the above cross

- A) A single gene product may produce more than one effect
- B) For the shape of seed gene shows dominant recessive relationship
- C) For starch synthesis gene B shows incomplete dominance
- D) It is an example for co-dominance

Which of the above inferences are true

Options :

- a) A,B,C,D
- b) A,B,C
- c) B,C
- d) B,C,D

32. A protein has been synthesized according to the sequence of nucleotides presented below.
AUGACGGCAGACAGAUGA

Identify the collect sequence of nucleotides of DNA that help in transcription of the protein.

Options:

- a) TAC TGC CGT TCT ACT
- b) ATG AGC GCA GCA GGT TGA
- c) UAC UGC CGU CUG UCU ACU
- d) ATG ACG GCA GAC AGT TGA

33. A scientist has selected a ds DNA molecule with $1700\text{\AA}$ length for his experiment and deleted 20 coils from the total DNA length. hi the remaining molecule 20° o of Adenine is present. The ratio among the following is

- A. Total no. of nucleotides deleted before conducting experiment
- B. No. of Guanins present in the experimented molecule
- C. No. of Thymines present in the experimented molecule

Options:

- a) 20:9:6
- b) 20:6:9
- c) 10:9:6
- d) 10:6:9

34. Fill up the blanks with suitable words.

Heavy DNA molecules synthesis by medium having A can be distinguished by normal DNA _____ B __process having C_ density gradient as D _____ is not a radioactive isotope.

Options:

	A	B	C	D
a)	^{15}N	Centrifugation	Cesium chloride	$^{15}\text{NH}_4\text{CL}$
b)	^{15}N	Cesium chloride	Centrifugation	$^{15}\text{NH}_4\text{CL}$
c)	$^{15}\text{NH}_4\text{CL}$	Centrifugation	Cesium chloride	^{15}N
d)	$^{15}\text{NH}_4\text{CL}$	Cesium chloride	Centrifugation	^{15}N

35. Find the correct statements

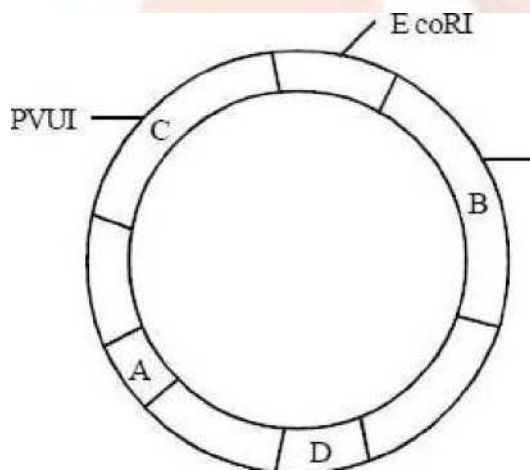
- A. In a chromosome a specific DNA sequence called 'Orit' where an alien DNA is linked and multiply
- B. To link the alien DNA vector needs single recognition sites for the restriction enzymes
- C. When cut by the restriction enzymes the resultant DNA fragments have different kind of sticky end. so that they can be joined together by DNA ligase.
- D. The definition for Biotechnology is the integration of

natural science and organism, cells, and molecular analogues for products and services

Options :

- a) A,B,C
- b) B,C
- c) A,B,D
- d) A,C,D

36. Observe the given P^{BR32}- vector model diagram and identify A. B. C, D coding regions respectively



Options :

- a) ori. amp^R. tet^R rop
- b) amp^R,tet^R,ori,rop
- c) ori,tet^R,amp^R,rop
- d) ori,rop,tet^R,amp^R

37. The following incorrect statement is

Options :

- a) Bt toxin is activated due to alkaline pH of insect gut.

- b) A and B nucleotide chains corresponding to human insulin are produced in E. coli through plasmid vector.
- c) Gene silencing is made by using sense and antisense DNA strands which bind to mRNA mRNA.
- d) In gene therapy, normal genes are inserted into persons cells or tissues to compensate non functional gene.

38. Choose the correct combination

I) Wheat	Himgni	High yielding
II) Wheat	Sonalka	Disease resistant
III) Flat bean	Pusa sem2	Aphid resistant
IV) Chilli	Pusa sada baliar	TMV resistant

Options;

- a) I,II,III
- b) I,III,IV
- c) II,III,IV
- d) I,II,IV

39. Identify the incorrect statements

- A. Trichodemia and baculovinis are biological control agents of plant pathogens
- B. The use of Biocontrol methods will greatly increase the dependance on toxic chemicals
- C. Methanogens helps in breakdown of cellulose for the generation of biogas
- D. Salinity tolerance and resistance to root home pathogens

cannot be associated with symbiotic fungal association
Options

- a) A,B
- b) C,D
- c) B,D
- d) A,D

40. Assertion (A): Dough used for making bread is fermented by using *Saccharomyces cerevisiae*.

Reason (R): *Saccharomyces cerevisiae* is called bakers yeast and brewers yeast

Options :

- a) (A) and (R) are correct and (R) is the correct explanation to (A)
- b) (A) and (R) are correct but (R) is not the correct explanation to (A)
- c) (A) is true. (R) is false
- d) (A) is false. (R) is true

Display Number Panel:

Yes

Group All Questions:

No

41. Assertion (A): Maintenance of relatively constant internal conditions which are different from that of surroundings is called homeostasis.

Reason (R): Living organisms maintain constant internal environment by various physiological adaptations.

Options :

- a) Both (A) and (R) are true. (R) is correct explanation for (A)
- b) Both (A) and (R) are true. (R) is not correct explanation for (A)
- c) (A) is true but (R) is wrong
- d) (A) is wrong but (R) is true

42. Assertion (A): Cardiac muscle is described as a functional syncytium.

Reason (R): The intercalated discs of heart muscles contain tight junctions which are responsible for rapid conduction of impulses.

Options :

- a) Both (A) and (R) are correct and (R) is the correct explanation to (A)
- b) Both (A) and (R) are correct but (R) is not correct explanation to (A)
- c) (A) is correct. (R) is incorrect
- d) (A) is wrong. (R) is correct

43. Read the following regarding cellular junctions and specializations

- A. Tight junctions prevent leakages of body fluids.
- B. Keratin filaments anchor desmosomes in inter cellular space.
- C. Communicating junctions comparable to plasma desmata of plant cells.
- D. Stereocilia are short and motile structures seen in internal ear.

Identify the correct statements

Options :

- a) A&B
- b) B&C
- c) A&C
- d) A&D

44. Match the following

Cells	Organism
A) Choanocytes	I) Pleurobrachia
B) Cnidoblasts	II) Adamsia
C) Flame cells	III) Lepisma
D) Lasso cells	IV) Sycon
	V) Schistosoma

The correct answer is

Options :

- | | | | | |
|----|----|----|----|-----|
| | A | B | C | D |
| a) | IV | II | V | III |
| | A | B | C | D |
| b) | V | II | I | IV |
| | A | B | C | D |
| c) | IV | II | V | I |
| | A | B | C | D |
| d) | IV | V | II | I |

45. Read the following statements

- A. Body cavity of Hirudinaria contains botryoidal tissue
- B. Tubifex is an oligochaete that lives in fresh water
- C. Clitellum is absent in unisexual annelids
- D. Larva of leeches is trochophore

Options :

- a) a,d
- b) a,b,c
- c) b,d
- d) b,c,d

46. In reptiles, this region of vertebral column contains only two vertebrae.

Options :

- a) Sacral
- b) Cervical
- c) Lumbar
- d) Caudal

47. Study the following

	Scientific Name	Common Name	Character
A	Torpedo	Electric ray	Ventral muscles are modified into electric organs

B	Pty as	Rat snake	Nonpoisonous snake
C	Struthio camelus	African ostrich	Syrinx is present
D	Belanoptera	Blue whale	Right systemic arch is present

Options :

- a) A&B
- b) B&D
- c) A&C
- d) 'B'

48. Match the following

Pseudopodium	Examples
A) Lobopodia	I) Elphidium
B) Reticnlopodia	II) Actinophrys
C) Filopodia	III) Polytoma
D) Heliopodia	IV) Entamoeba
	V) Englypia

The correct answer is

- | | | | |
|-------|----|---|-----|
| A | B | C | D |
| a) IV | I | V | II |
| b) II | V | I | IV |
| c) IV | I | V | III |
| d) IV | II | V | I |

49. The following drugs are used in sleeping pills

Options :

- a) Barbiturates
- b) Amphetamines
- c) Benzodiazepines
- d) Cannabinoids

50. Assertion (A): Excessive doses of cocaine cause hallucinations

Reason (R): It has a potent stimulating action on central nervous system

Options :

- a) Both (A) and (R) are true. (R) is correct explanation for (A)
- b) Both (A) and (R) are true. (R) is not the correct explanation for (A)
- c) (A) is true. But (R) is false
- d) (A) is false. But (R) is true.

51. In erythrocytic cycle OF Plasmodium vivax. the following stages are observed.

- A. Rosette stage
- B. Trophozoites
- C. Amoeboid stage
- D. Signet ring stage
- E. Liberation of erythrocytic merozoites
- F. Shuffner's dots in RBC

Arrange them in correct sequence

Options :

- a) b,d,f,c,a,e
- b) b,d,c,f,a,e
- c) e,a,f,c,d,b
- d) b,d,c,f,e,a

52. Read the following statements

- A. Cocaine interferes with neurotransmitter dopamine and causes depression
- B. Nicotine raises the blood pressure and heart rate
- C. Cannabinoids show their effects on cardiovascular system
- D. Morphine acts as a sedative and pain killer

Correct statements are

Options :

- a) B & C
- b) B,C&D
- c) A, B & D
- d) A, C & D

53. Identify the Incorrect statement regarding cockroach

Options :

- a) The claws and the arolium help in locomotion on rough

surfaces

- b) Plantulae are useful in locomotion on smooth surfaces which are located on tibia of less
- c) Mycetocytes contain symbiotic bacteria
- d) Stomodeal valve secretes peritrophic membrane

54. Match the following

Cells of fat body	Function
A) Trophocytes	I) Stole uric acid
B) Mycetocytes	II) Synthesis and storage of lipids
C) Oenocytes	III) Storage of urea
D) Urate cells	IV) Storage of food
	V) Store syniboitic bacteria

The correct answer is

A	B	C	D
a) III	IV	V	I
b) IV	V	III	II
c) IV	III	V	I
d) IV	V	II	I

55. Choose the incorrect combination about nitrogen fixation

Options:

- a) Ammonia $\longrightarrow$ Nitrite - Nitrosomonas
- b) Nitrite $\longrightarrow$ Nitrate - Nitrobacter
- c) Denitrification - Thiobacillus

d) Ammonification - Ninosococcus

56. Assertion (A): Cyclomorphosis is a seasonal adaptation in certain animals to stabilize the movement in water currents

Reason (R): Typical morphs can resist the water currents better to stay in the water rich hi food materials

Options :

- a) Both (A) and (R) are collect, and (R) is the correct explanation to (A)
- b) Both (A) and (R) are collect, but (R) is not correct explanation to (A)
- c) (A) is correct. (R) is not correct
- d) (A) is not correct (R) is correct

57. Study the following and pick up the wrong one

Options:

- a) White matter of central nervous system contains myelinated neurons
- b) The narrow gap between the terminal fiber of axon and dendrite of next neuron is called synopsis
- c) Ependymal cells keep the cerebrospinal fluid in circulation
- d) A group of nerve cell bodies in central nervous system is called nucleus

58. Study the following statements about adaptations of animals

- A. Protopterus shows anadromous migration to survive in summer
- B. Photoperiodism is the influence of light on velocity of movement
- C. Kangaroo rat uses metabolic water to survive on land
- D. "Mammals from cold climates have short extremities of body" is Allen's rule

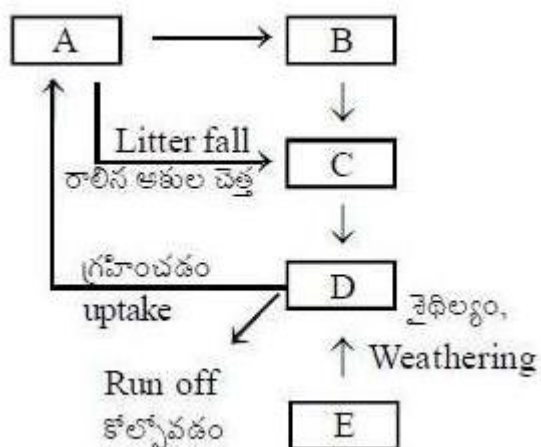
From the above, identify the correct statements

Options:

- a) A, C
- b) B, C
- c) B, D
- d) C, D

59. Study the model of phosphorus cycle

- I. Rock Minerals
- II. Producers
- III. Detritus
- IV. Soil solution
- V. Consumers



Identify the A. B.C. D and E

Options :

- a) A-IV B-V C-III D-II E-1
- b) A-II B-IV C-I D-V E-1
- c) A-II B-V C-III D-IV E-1
- d) A-III B-V C-I D-II E-IV

60. Sphincter of Oddi regulates the flow of

Options :

- a) Chyme from pyloric part of stomach into duodenum
- b) Pancreatic juice from pancreas into duodenum
- c) Succus entericus from ileum into duodenum
- d) Pancreatic juice and bile juice into duodenum

61. Arrange the following in order of increasing volume

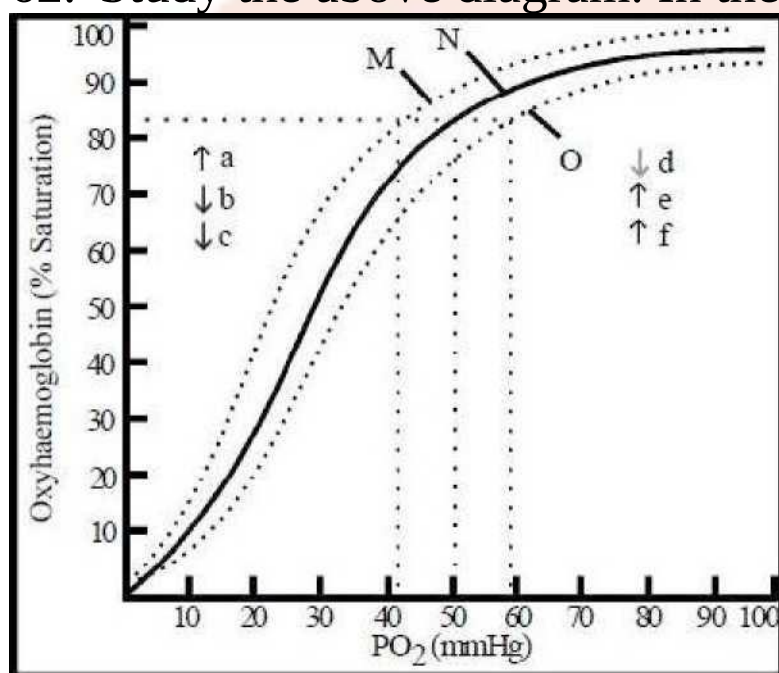
- a) Tidal volume
- b) Residual volume
- c) Expiratory reserve volume

d) Vital capacity

Options

- a) d,c,b,a
- b) d,b,c,a
- c) b,c,d,a
- d) c,b,d,a

62. Study the above diagram. In the diagram



N = Oxygen haemoglobin dissociation curve

M = Left side shift of the curve

O = Right side shift of the curve Identify the factors a, b, c, d, e and f in the graph.

Options :

- a) a,d=CO₂ : b,e = Temperature : c,f=pH
- b) a,d = CO₂: b,e = pH : c,f= Temperature
- c) a,d = O₂ : b,e = Temperature : c,f=pH
- d) a,d=pH : b,e = CO₂ : c,f= temperature

63. Assertion (A): Ischemia is a phenomenon of decrease in supply of blood to the heart muscles

Reason (R): Narrowing of blood vessels to the heart leads to ischemia

Options :

- a) Both (A) and (R) are correct and (R) is the correct explanation to (A)
- b) Both (A) and (R) are correct but (R) is not correct explanation to (A)
- c) (A) is correct. (R) is not correct
- d) (A) is not correct but (R) is correct

64. Study the following and choose the correct combinations

	Column - I	Column-II	Column - III
A	Factor VII	Release of tissue thromboplastin	Formation of thrombin
B	Warfarin	Antagonistic to Vit-K	Prevents synthesis of Christmas factor
C	Atrial systole	Lasts about 0.1 Sec.	Closure of AV valves
D	Factor XII	Hageman's factor	Formation of prothrombin activator

Options :

- a) A,C

- b) B,C
- c) A,D
- d) B,D

65. Read the following statements

- A. Counter current mechanism helps to concentrate urine
- B. Creatinine is an threshold substance
- C. The descending limb of loop of Henle is permeable to Na^+
- D. The ascending limb of loop of Henle is permeable to water

Choose the correct statements

Options :

- a) A&B
- b) B&C
- c) C&D
- d) A&D

66. The joint "Epiphyseal plate¹" present between Epiphysis and Diaphysis is a

Options :

- a) Syndesmoses
- b) Gomphose
- c) Synchondrose
- d) Symphysis

67. Study the following events about mechanism of hearing

- A. Bending of hair cells of basilar membrane
- B. Generation of waves in fluids of cochlea
- C. Pressing of tectorial membrane
- D. Vibrations of tympanum
- E. Ripple in basilar membrane

Arrange the above sequence

Options :

- a) D-E-A-B-C
- b) D-B-C-E-A
- c) D-B-E-A-C
- d) D-B-A-E-C

68. Match the following

List -1	List - II
A) Oogenesis	I) Oxytocin
B) Ovulation	II) Melanocyte stimulating hormone
C) Parturition	III) Prolactin
D) Lactation	IV) Follicle stimulating hormone
	V) Luteinising hormone

The correct answer is

A B C D

- a) II I IV V
- b) II I III IV
- c) IV V I II
- d) IV V I III

69. Study the following

- A. In secondary lymphoid organs The immuno competent T-cells are transformed into cytotoxic T-cells and natural killer cells
 - B. Immunity developed by the antibodies that are released into the fluids of the body is called humoral immunity
 - C. After getting into the body of person. HIV enters the macrophages and dendritic cells
 - D. Bursa Fabricius of birds is a secondary lymphoid organ
- Among the above, the wrong statements are

Options :

- a) A,B
- b) B,C
- c) C,D
- d) A,D

70. The placenta of human beings is called

Options :

- a) Chorio allantoic. Discoidal, Haemochorial Deciduate placenta
- b) Chorioallantoic, Metadiscoidal, Haemoendo chorial Non- deciduate placenta

- c) Yolk sac, Metadiscoidal, Haemochorial, Deciduate placenta
- d) Chlorio allantoic. Diffused, Haemochorial Non-deciduate placenta

71. Study the following and choose the correct combination about human reproduction

	List -1	List - II	List - III
(A)	Trophoblast	Proteolytic enzymes	Implantation
(B)	Acrosin	Acrosome reaction	Corona penetrating enzyme
(C)	Secondary spermatocyte	Meiosis - H	Large spermatid and polar body
(D)	Placenta	Somatotrophin	Increased availability of glucose to foetus

Options :

- a) A,C
- b) B,C
- c) B,D
- d) A,D

72. Read the following and identify the mismatch pair
Options :

- a) Capacitation - Changes in spermatozoa

- b) Compaction - Conti action of ova
- c) Menarche - The first menstruation
- d) Amphimixis - Fusion of gametic pro nuclei

73. Karyotype of a Drosophila is AAA+XX, it is

Options :

- a) Female
- b) Male
- c) Intersex
- d) Meta female

74. Assertion (A): In moths the sex of the offspring depends on the type of ovum that is fertilized.

Reason (R): In moths the females are heterogametic.

Options :

- a) Both (A) and (R) are true. R is the correct explanation for (A)
- b) Both (A) and (R) are true. R is not a correct explanation for (A)
- c) (A) is true but (R) is false
- d) (A) is false but (R) is true

75. Study the following statements about genetic disorders

A. Phenylketonuria is caused by a mutation in the gene located on chromosome 12

B. In Cooley's Anaemia the beta chain binds to RBC and damage them

- C. Cystic fibrosis causes increased resorption of Na and H_2O from respiratory epithelium
- D. Individuals heterozygous to sickle cell trait always lead healthy life

From the above identify the correct statements

Options :

- a) A,D
- b) B,C
- c) A,C
- d) B,D

76. In a population with two alleles 'B' and 'b' having allele frequency 0.7 and 0.3. In Hardy-weinberg equilibrium how many individuals in a sample of 250 can be expected to be heterozygotes (Bb).

Options :

- a) 52
- b) 105
- c) 21
- d) 42

77. Assertion (A): Oversized antlers in the Irish deer lead to its extinction

Reason (R): Darwinism explains the origin of new species is due to accumulation of adaptive variations

Options:

- a) Both (A) and (R) are true (R) is the collect explanation to (A)
- b) Both (A) and (R) are true. (R) is not a correct explanation to (A)
- c) (A) is true but (R) is false
- d) (A) is false but (R) is true

78. The biomedical technique used to analyse skull fractures

Options:

- a) EEG
- b) MRI Scan
- c) PET Scan
- d) CAT Scan

79. Match the following

List -1	List - II
A) Shortened Q-T interval	I) Hypokalemia
B) Livened T wave	II) Tachycardia
C) Elevated S-T segment	III) Hyperkalemia
D) Shortened P-R interval	IV) Hypercalcemia
	V) Myocardial infarction

The correct answer is

- | | A | B | C | D |
|----|----|---|-----|-----|
| a) | IV | I | III | II |
| b) | IV | I | V | III |
| c) | IV | I | V | II |

d) IV III II V

80. Read the following statements regarding cancer

- A. Retinoblastoma protein is a tumor suppressor protein
- B. P53 protein allows the cell to enter the apoptosis when DNA damage is irreparable
- C. Proto-oncogene mutations are recessive
- D. P53 protein plays key role in regulating G2 check point

From the above. Identify the correct statements

Options :

- a) A&B
- b) B&C
- c) B,C&D
- d) A,B&C

81. If P,Q and R are physical quantities having different dimensions, then the following combination never be a meaning full quantity

Options :

- a) $\frac{PQ}{R}$
- b) $\frac{R}{PQ}$
- c) $\frac{Q^2}{PR}$

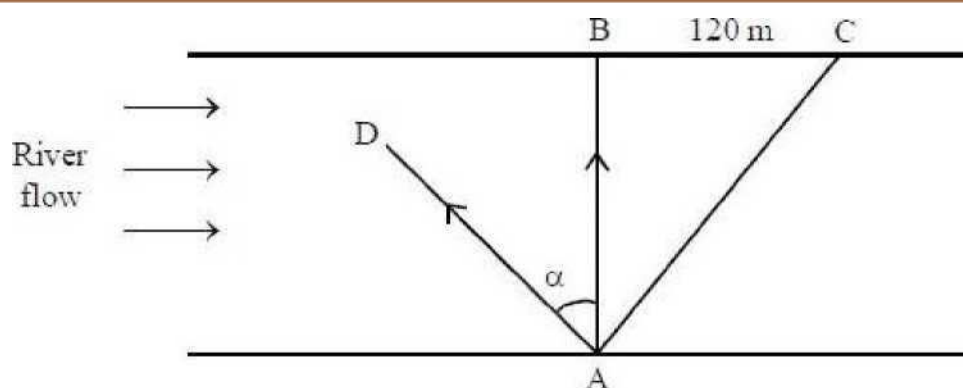
d) $\frac{PQ}{R+Q}$

82. Two trains P and Q each of length 400 m are moving along parallel tracks, in the same direction with a same uniform speed of 20 ms^{-1} . Driver of train P decides to overtake train Q and accelerates his train by 1 ms^{-2} . If the train P crosses the train Q in 50 s. then the initial distance between the two trains was

Options :

- a) 450 m
- b) 800 m
- c) 540 m
- d) 425 m

83. If a boat moves perpendicular to the bank of a river from a point A. it reaches a point C' which is at 120 m down the stream from a point B on the opposite bank in 10 minutes as shown in the figure. If the boat moves with the same velocity at an angle α along AD upstream it reaches the point B in 12.5 minutes. The width of the river and the angle α respectively are.
(Line AB is perpendicular to the banks of the river).



- a) $100 \text{ m} \cdot \sin^{-1}(0.8)$
- b) $200 \text{ m} \cos^{-1}(0.8)$
- c) $300 \text{ m} \cdot \tan^{-1}(0.6)$
- d) $300 \text{ m} \cos^{-1}(0.6)$

84. Two forces simultaneously act on a particle by making an angle of 120° with each other. If one of them is reversed, the acceleration of the particle becomes VJ times. Then the ratio of magnitude of forces is

Options :

- a) 1 : 1
- b) 1 : 4
- c) 1 : 2
- d) 2 : 1

85. A body starts with a velocity 'u' from the top of an inclined plane of length 'L' and an angle of inclination 30° . The upper part of the incline of length $\frac{2L}{3}$ is rough and the remaining part is smooth. If the body slides down and reaches the bottom with the same velocity Tf . then the coefficient of kinetic friction of the rough part of the

incline is

Options :

- a) $\frac{1}{\sqrt{3}}$
- b) $\frac{1}{\sqrt{2}}$
- c) $\frac{\sqrt{3}}{2}$
- d) $\frac{2}{\sqrt{3}}$

86. All object is placed at the top of a frictionless hemispherical metallic surface of radius 3m.If the object is just disturbed, it will slide down and leaves the hemispherical surface at a height of

Options:

- a) 1 m
- b) 2 m
- c) 3 m
- d) 4 m

87. A force (F) acting oil a body varies with the velocity (v) of the body as $F = \frac{C}{v^2}$ where C' is a constant. Then the work done by the force in 't' seconds is proportional to

Options :

- a) $\frac{t}{v}$
- b) $\frac{1}{vt}$
- c) $\frac{v}{t}$
- d) vt

88. A projectile is fired on a horizontal ground. The coefficient of restitution between the projectile and the ground is 'e'. The ratio of horizontal ranges in first two collisions with the ground is

Options :

- a) $\frac{1}{e}$
- b) $\frac{1}{e^2}$
- c) e
- d) e^2

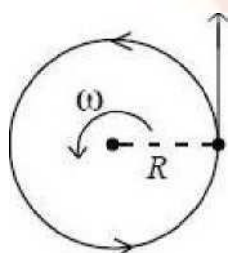
89. The center of mass of a system of three particles of masses 1 g, 2g and 3g is at the origin of a co-ordinate system. If a particle of mass 4g and having position vector $\alpha(\hat{i} + 2\hat{j} + 3\hat{k})$ is added to the three-particle system, then the center of mass of the four-particle system

becomes (1.2.3). The value of 'cz' is

Options:

- a) $\frac{10}{3}$
- b) $\frac{5}{2}$
- c) $\frac{1}{2}$
- d) $\frac{2}{5}$

90. A girl with mass 'm' is standing at the edge of a merry-go-round having moment of inertia T. radius R and angular velocity ' ω ' as shown in figure. The girl jumps off the edge of the merry-go-round with a velocity ' v ' with respect to the ground in a direction tangent to the edge. The new angular velocity of the merry-go-round is



Options :

- a) $\left[\frac{I\omega^2 - mv^2}{I} \right]^{\frac{1}{2}}$
- b) $\left[\frac{(I + mR^2)\omega^2 - mv^2}{I} \right]^{\frac{1}{2}}$
- c) $\left[\frac{I\omega - mvR}{I} \right]$
- d) $\left[\frac{(I + mR^2)\omega - mvR}{I} \right]$

91. The potential energy of a particle of mass 5 mg executing simple harmonic motion along x-axis is given by $V(x) = (1.23) + (4.84) x^2$. where $V(x)$ is in joules and x is in meters. The frequency of oscillation of the particle is _____ Hz.

Options :

- a) 77
- b) 7
- c) 22
- d) 2.2

92. The period of revolution of an earth satellite close to the surface of earth is 90 minutes. The time period of another satellite in an orbit at a distance of three times the radius of earth from the earth's surface will be

Options :

- a) $90\sqrt{8}$ minutes

- b) 360 minutes
- c) 720 minutes
- d) 270 minutes

93. A simple pendulum with a uniform metal wire as the string has a time period T_1 . Young's modulus, length and area of cross-section of the wire are respectively Y , l and A . When an additional mass 'in' is added to the bob, the time period becomes T_2 . Then $\frac{1}{y}$ is equal to

Options:

- a) $\frac{2mgR}{3k}$
- b) $\left[\left(\frac{T_2}{T_1} \right)^2 + 1 \right] \frac{A}{mg}$
- c) $\left[1 + \left(\frac{T_2}{T_1} \right) \right] \frac{A}{mg}$
- d) $\left[\left(\frac{T_2}{T_1} \right)^2 - 1 \right] \frac{A}{mg}$

94. A soap has bubble radius 'R' and thickness d ($\ll R$). It collapses into a spherical drop. The ratio of excess pressure in the drop to the excess pressure inside the bubble is

Options :

a) $\frac{\lambda(\pi - 2\phi)}{4\pi}$

b) $\left(\frac{R}{8d}\right)^{\frac{1}{3}}$

c) $\left(\frac{R}{6d}\right)^{\frac{1}{3}}$

d) $\left(\frac{R}{24d}\right)^{\frac{1}{3}}$

95. A metallic pendulum clock shows 4 s more each day at 5 °C in winter and 12 s less at a temperature 45 °C in summer. Then the coefficient of linear expansion of the metal in terms of $10^{-6} \text{ }^{\circ}\text{C}^{-1}$

Options :

a) 18.4

b) 4.6

c) 27.6

d) 9.2

96. Ice starts forming in a lake with water at 0 °C. when the atmospheric temperature is -10 °C. If the time taken for 1 cm of ice to be formed is 7 hours, the time taken for the thickness of ice to change from 1 cm to 2 cm is (in hours)

Options :

- a) 3.5
- b) 7
- c) 14
- d) 21

97. The specific heat (c) of a metal at very low temperatures (T) is given by $c = aT + bT^3$ where $a = 0.01 \text{ J kg K}^{-2}$ and $b = 16 \times 10^{-4} \text{ J kg K}^{-4}$. The amount of heat energy required to increase the temperature of 2 kg of the metal from 1 K to 3 K is _____ mJ.

Options :

- a) 144
- b) 72
- c) 288
- d) 216

98. An ideal monoatomic gas is confined in a cylinder by a spring-loaded piston of area of cross-section $8 \times 10^{-3} \text{ m}^2$. Initially, the gas is at 300 K and occupies a volume of $2.4 \times 10^{-3} \text{ m}^3$ and the spring is in its relaxed state as shown in the figure. The gas is heated by a small electric heater until the piston moves out slowly by 0.1 m. The final temperature of the gas is

The force constant of the spring = 8000 Nm^{-1}

Atmospheric pressure = $1 \times 10^5 \text{ Nm}^{-2}$

The piston is massless and there is no friction between the piston and cylinder)

Options :

- a) 800 K
- b) 400 K
- c) 600 K
- d) 300 K

99. The absolute temperature at which hydrogen molecules escape from a planet's surface is 'm' is mass of hydrogen molecules, 'k' is Boltzmann constant. 'R' is radius of the planet, 'g' is the acceleration due to gravity on the planet)

Options :

- a) $\frac{mgR}{2k}$
- b) $\frac{2mgR}{k}$
- c) $\frac{3mgR}{2k}$
- d) $\frac{2mgR}{3k}$

100. The fundamental frequency of a uniform wire under tension is in resonance with the fundamental frequency of closed tube. If the tension in the wire is increased by 8 N. it is in resonance with first overtone of the same

closed tube. The initial tension in the wire is

Options :

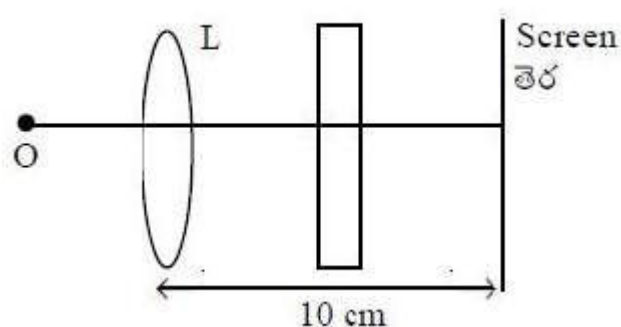
- a) 0.5 N
- b) 1 N
- c) 2 N
- d) 4 N

101. The path difference between the two waves represented

by $x_1 = 4\sin\left(\omega t - \frac{2\pi x}{\lambda}\right)$ and $x_2 = 4\sin\left(\omega t - \frac{2\pi x}{\lambda} - \phi\right)$

- a) $\frac{\lambda(2\pi - \phi)}{4\pi}$
- b) $\frac{\lambda(\pi - 2\phi)}{2\pi}$
- c) $\frac{\lambda(\pi - \phi)}{4\pi}$
- d) $\frac{\lambda(\pi - 2\phi)}{4\pi}$

102. A convex lens of focal length 9 cm is placed 10 cm before a screen. A glass plate ($\mu = 1.5$) of thickness 1.5 cm is inserted between the lens and the screen as shown in the figure. To focus image of an object 'O' on the screen, the object must be placed at a distance of x cm from the lens. Then the value of x is



Options :

- a) 85 cm
- b) 117 cm
- c) 171 cm
- d) 42.5 cm

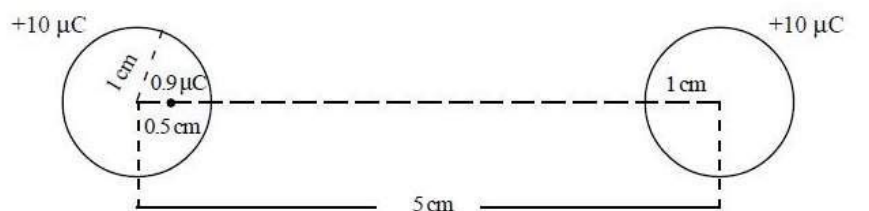
103. In Young's double slit experiment, intensity at a point on the screen is $\frac{1}{4}$ th of the maximum intensity. Then the angular position of this point is (λ - wavelength of light, d - separation between the two slits)

Options:

- a) $\sin^{-1}\left(\frac{\lambda}{d}\right)$
- b) $\sin^{-1}\left(\frac{\lambda}{3d}\right)$
- c) $\sin^{-1}\left(\frac{3\lambda}{2d}\right)$
- d) $\tan^{-1}\left(\frac{\lambda}{d}\right)$

104. Two non-conducting thin spherical shells of uniformly distributed charge of $+10 \mu\text{C}$ each and radius 1 cm each are placed with a separation of 5 cm between their centres. A positive charge of $0.9 \mu\text{C}$ is placed inside one of the shells at a distance of 0.5 cm from its centre on the line joining the centres of the two shells as shown in the figure. The net electrostatic force on $0.9 \mu\text{C}$ charge is

— $N \left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2} \right)$



Options :

- a) 10
- b) 20
- c) 30
- d) 40

105. The bob of a pendulum of mass 50 g with a positive charge $5 \mu\text{C}$ is oscillating with a time period T_1 . An electric field of intensity $6.4 \times 10^4 \text{ NC}^{-1}$ is applied vertically upwards. Now the time period is T_2 , then the value of $\frac{T_2}{T_1}$ is ——— ($g = 10 \text{ ms}^{-2}$)

Options :

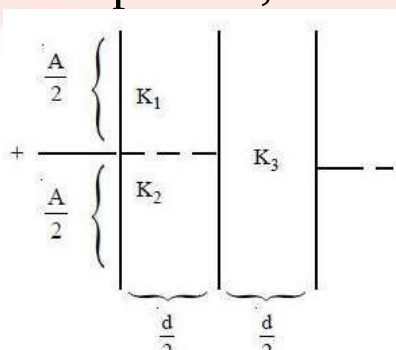
- a) 2.67
- b) 1.67
- c) 4.7
- d) 6.67

106. Two electric dipoles each of moment $5 \times 10^{-12} \text{ Cm}$ form a symbol '+' with their axes along the co-ordinate axes. The electric potential at a point 20 cm away in a direction making an angle of 30° with axis is

Options :

- a) 0.56 V
- b) 4.56 V
- c) 3 V
- d) 1.536 V

107. A parallel plate air capacitor with plate area 'A' and plate separation 'd' filled with three dielectric materials of dielectric constants K_1, K_2 and K_3 is as shown in the figure. If the three dielectric materials are replaced by a single dielectric material of dielectric constant 'K' filling the entire capacitor, the capacity remains the

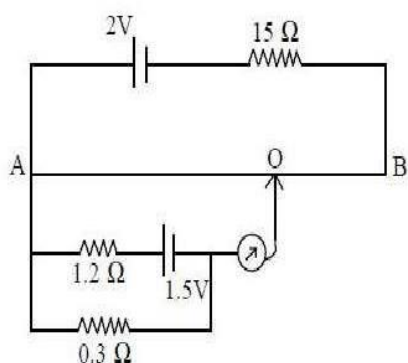


same, then

Options :

- a) $\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2} + \frac{1}{2K_3}$
- b) $\frac{1}{K} = \frac{1}{K_1 + K_2} + \frac{1}{2K_3}$
- c) $\frac{1}{K} = \frac{K_1 K_2}{K_1 + K_2} + 2K_3$
- d) $K = \frac{K_1 K_2}{K_1 + K_2}$

108. AB is a uniform wire of length 1 m and resistance of $10\ \Omega$ as shown in figure. The length 'AO' when galvanometer shows no deflection is



Options :

- a) 75 cm
- b) 30 cm
- c) 18.75 cm
- d) 37.5 cm

109. A battery of emf $8\ V$ and internal resistance $0.5\ \Omega$ is being charged by a $120\ V$ d.c. supply using a series resistor of $15.5\ \Omega$. The terminal voltage of $8\ V$ battery during charging is

Options :

- a) 11.5 V
- b) 112V
- c) 14 V
- d) 6V

110. In Bohr model of Hydrogen atom, the electron revolves around the nucleus in a path of radius $5 \times 10^{-11} \text{ m}$ at a frequency of $5 \times 10^{15} \text{ rev/s}$. The equivalent dipole moment is

Options:

- a) $6.28 \times 10^{-24} \text{ Am}^2$
- b) $6.28 \times 10^{-26} \text{ Am}^2$
- c) $62.8 \times 10^{-24} \text{ Am}^2$
- d) $0.628 \times 10^{-24} \text{ Am}^2$

111. The force acting on a straight wire of length 10 cm carrying a current of 4 A in the +Z direction due to a uniform magnetic field is $(-2\bar{i} + 2\bar{j}) \text{ N}$. If the wire is rotated so that it carries a current in +ve X-direction. the force on the wire becomes $(-2\sqrt{2}\bar{j} + 2\bar{k}) \text{ N}$. The strength of the magnetic field is _____ T.

Options :

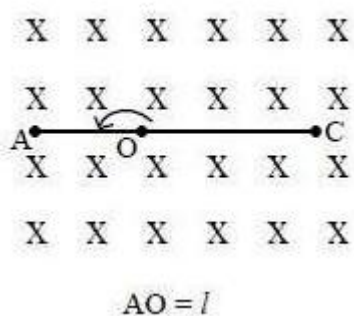
- a) 10
- b) 15
- c) 18
- d) 20

112. A bar magnet has coersivity of $6 \times 10^3 \text{ Am}^{-1}$. It is desired to demagnetize it by inserting it inside a solenoid of length 15 cm and having 90 turns. The current that should be sent through the solenoid is

Options :

- a) 10A
- b) 5A
- c) 20A
- d) 15A

113. A uniform magnetic field of induction 'B' is perpendicular to the plane of paper and into the paper. A conducting rod AC of length '4l' is rotating about a point 'O' with a uniform angular velocity ' ω ' in this field as shown in the figure. If $AO = l$, then the potential difference between the points 'O' and 'C' is



Options :

- a) $\frac{Bl^2\omega}{2}$
- b) $\frac{9Bl^2\omega}{2}$
- c) $Bl^2\omega$

d) $\frac{9Bl^2\omega}{4}$

114. A capacitor of capacitance $4 \mu\text{F}$ is charged to 100 V. The charging battery is then disconnected and a coil of inductance 10 mH is connected across it. [Resistance of the coil is small and negligible]. The peak value of current in the circuit is

Options :

- a) 1A
- b) 2A
- c) 3A
- d) 4A

115. Match the type of electromagnetic wave given in list A with its use given in list B

List A	List B
a) Infrared	i) Air craft navigation
b) Ultraviolet	ii) Treatment of certain forms of cancer
c) X-Rays	iii) Remote switches
d) Microwave	iv) Lamps to kill genus in water purifiers

Options :

- | | | | | |
|----|----|----|-----|-----|
| | a | b | c | d |
| a) | i | ii | iv | iii |
| b) | ii | i | iii | iv |

- c) iii iv ii i
d) iv iii i ii

116. Assertion (A): Threshold frequency of a photo metal depends on the intensity of incident light.

Reason (R): Greater is the intensity of incident light, greater is the photoelectric current.

Options :

- a) Both (A) and (R) are true and (R) is the correct explanation of (A)
- b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
- c) (A) is true, but (R) is not true
- d) (A) is not true. but (R) is true

117. The radiation emitted when an electron jumps from an orbit of $n = 2$ to an orbit of $n = 1$ in Hydrogen atom is made to fall on a metal surface with work function 1.2 eV. The maximum velocity of photoelectrons emitted is nearly

Options :

- a) $6 \times 10^5 \text{ ms}^{-1}$
- b) $3 \times 10^5 \text{ ms}^{-1}$
- c) $2 \times 10^5 \text{ ms}^{-1}$
- d) $8\sqrt{5} \times 10^5 \text{ ms}^{-1}$

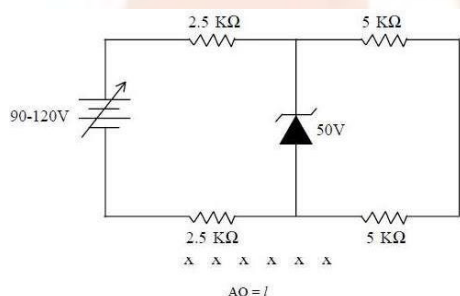
118. The half-life of a radioactive isotope 'X' is 20 years. It decays to another stable element 'Y' The number of

atoms in 'X' and 'Y' are found to be in the ratio 1:7 in a rock. The age of the rock is

Options :

- a) 40 years
- b) 60 years
- c) 80 years
- d) 100 years

119. The ratio of minimum and maximum currents through the zener diode in the given circuit is



Options :

- a) $\frac{1}{3}$
- b) $\frac{8}{9}$
- c) $\frac{2}{5}$
- d) $\frac{3}{7}$

120. An audio signal represented by $40\sin 3\pi(4000\tau)$ is amplitude modulated by carrier wave $90\sin 3\pi(9000\tau)$.

The percentage of modulation index of the modulated signal is

Options

- a) 4.44%
- b) 3.33%
- c) 33.3%
- d) 44.4%

Display Number Panel: Yes
Group All Questions: No

121. The radius (in nm) of first orbit of He^+ is:

Options :

- a) 0.0529
- b) 0.02645
- c) 0.01322
- d) 0.529

122. The kinetic energy of an electron is $4.55 \times 10^{-25} \text{ J}$. What is its wavelength (in nm)? (mass of electron $= 9.1 \times 10^{-31} \text{ kg}$)

Options :

- a) 728
- b) 850
- c) 896
- d) 786

123. Which of the following electronic configurations represent the most metallic element?

Options :

- a) $[Ne]3s^2 3p^5$
- b) $[Xe]6s^1$
- c) $[Xe]6s^2$
- d) $[Xe]4f^{14} 5d^{10} 6s^2 6p^1$

124. In which of the following set all molecules have equal bond order?

Options :

- a) H_2, He_2, B_2
- b) C_2, Li_2, O_2
- c) H_2, Li_2, B_2
- d) O_2, H_2, C_2

125. The sum of number of c bonds and non bonding electron pairs is maximum in

Options :

- a) CO_2
- b) Cl_2
- c) CH_4
- d) C_2H_4

126. 12 L of SO_2 gas diffused through a porous membrane in 20 minutes. Under same conditions 36 L of $\underline{\text{X}}$ gas diffused in 30 minutes. The molar mass of the gas ' $\underline{\text{X}}$ ' in g mol^{-1} is:

Options :

- a) 64
- b) 32
- c) 16
- d) 8

127. In an organic compound containing C and H. the % C is 3 times to that of % H. What is the compound?

Options :

- a) C_6H_6
- b) C_6H_{12}
- c) C_2H_6
- d) CH_4

128. When one mole of manganese is burnt completely to form $\text{Mn}_3\text{O}_{4(s)}$ at standard state conditions, the amount of heat liberated is $\Delta H_f^0 \text{ of } \text{Mn}_3\text{O}_4 = -1388 \text{ kJ mol}^{-1}$

Options :

- a) 624.67 kJ
- b) 462.67 kJ
- c) 476.67 kJ

d) 426.67 kJ

129. At 298 K, the equilibrium constant for a reaction is $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$ 100. If the initial concentration of each species is 1 M. the equilibrium concentration of D in mol L^{-1} is

Options :

- a) 0.818
- b) 1.818
- c) 1.182
- d) 0.182

130. At 27°C if the solubility product of $M(OH)_2$ is 3.2×10^{-11} , its solubility in g L^{-1} is (molar mass of $M(OH)_2$ is 58 g mol^{-1})

Options :

- a) 2×10^{-4}
- b) 1.16×10^{-14}
- c) 1.16×10^{-2}
- d) 5.8×10^{-3}

131. The number of hydrogen-bonded and co-ordinated water molecules present in hydrated copper sulphate salt are respectively

Options :

- a) 1,4

- b) 2,3
- c) 4,1
- d) 0,5

132. Identify the correct statements from the following

Options :

- a) Nitrates of all alkali metals give NO_2 and O_2 on heating
- b) Except LiHCO_3 , all the hydrogen carbonates of alkali metals are obtained in solid form
- c) Except LiCl , all the chlorides of alkali metals are deliquescent
- d) Carbonates of all alkali metals give CO_2 on thermal decomposition

133. In water orthoboric acid behaves as a weak monobasic acid giving H_3O^+ and

Options :

- a) $[\text{B}(\text{OH})_4]^+$
- b) H_2BO_2^-
- c) $[\text{B}(\text{OH})_4]^-$
- d) H_2BO_2^+

134. The gas common to both producer gas and water gas is

Options :

- a) N_2
- b) H_2

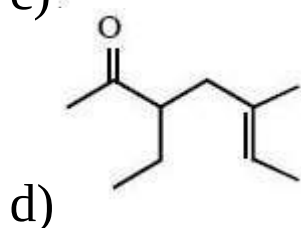
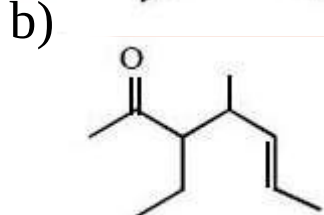
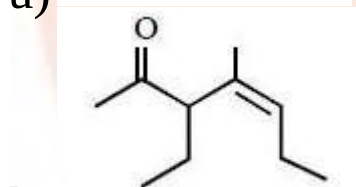
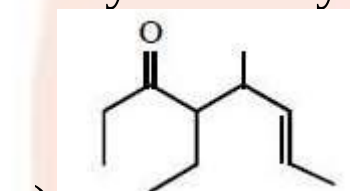
- c) CO
- d) SO₂

135. The concentrations of nitrate ion in drinking water exceeds 500ppm

Options :

- a) Kidney damage
- b) Laxative effect
- c) Methemoglobinemia
- d) Liver damage

136. Which one of the following structures represent 3-ethyl-4-methylhept-5-en-2-one?



137. In the addition of bromine to ethene, the intermediate formed is:

Options :

- a) Carbocation
- b) Cyclic bromonium ion
- c) Free radical
- d) Carbanion

138. The group which deactivates the benzene ring towards the electrophilic but directs the incoming group principally to o - and p - positions is

Options:

- a) -NH_2
- b) -NO
- c) $\text{-C}_2\text{H}_5$
- d) -Cl

139. Which one of the following relationships is collect for cubic (C) and rhombohedral (R) unit cells?

Options :

- | <u>C</u> | <u>R</u> |
|---|---|
| a) $a \neq b \neq c; \alpha = \beta = \gamma$ | $a = b = c; \alpha = \beta = \gamma = 90^\circ$ |
| <u>C</u> | <u>R</u> |
| b) $a = b = c; \alpha \neq \beta \neq \gamma$ | $a \neq b \neq c; \alpha = \beta = \gamma = 90^\circ$ |

c) $\frac{C}{R}$
 $a = b = c; \alpha = \beta = \gamma = 90^\circ$

$a = b \neq c; \alpha = \beta = \gamma = 90^\circ; \gamma = 120$

d) $\frac{C}{R}$
 $a = b = c; \alpha = \beta = \gamma = 90^\circ$ $a = b = c; \alpha = \beta = \gamma \neq 90^\circ$

140. Which one of the following is not correct?

Options :

- a) For an ideal solution $\Delta_{mix}H = 0; \Delta_{mix}V = 0$
- b) A mixture of phenol and aniline shows negative deviation in its vapour pressure
- c) A mixture of carbondisulphide and acetone shows positive deviation in its vapor pressure
- d) Nitric acid and water mixture shows large positive deviation in its vapor pressure and forms minimum boiling point azeotrope

141. An aqueous solution freezes at -0.372°C . If K_f and K_b for water are $1.86 \text{ K kg mol}^{-1}$ and $0.53 \text{ K kg mol}^{-1}$ respectively, the elevation in boiling point of same solution in K is:

Options :

- a) 0.72
- b) 0.46

- c) 4.6
- d) 0.106

142. The specific conductance of 0.01 M weak acid HA is $2.0 \times 10^{-4} \text{ S cm}^{-1}$. If its degree of dissociation is 0.05. its limiting molar conductivity in $\text{S cm}^2 \text{ mol}^{-1}$ is:

Options :

- a) 400
- b) 300
- c) 200
- d) 40

143. Assertion(A): Half life period of first order reaction independent of concentration of the reactant.

Reason (R): 99.9% reaction of first order reaction completes in 10 half lives.

The correct answer is:

Options :

- a) (A) and (R) are correct, and (R) is the correct explanation of (A)
- b) (A) and (R) are correct, but (R) is not the correct explanation of (A)
- c) (A) is correct but (R) is not correct
- d) (A) is not correct but (R) is correct

144. Which one of the following statements is not correct?

Options:

- a) Plant latex is a colloidal solution of rubber particles which are negatively charged
- b) Hemoglobin is a positively charged sol
- c) On prolonged dialysis the colloid is stabilized
- d) The process of setting of colloidal particles is coagulation

145. The method of zone refining of metals is based on the principle of

Options :

- a) Greater mobility of the pure metal than that of the impurity
- b) Greater solubility of the impurity in the molten state than in the solid of metal
- c) Higher melting point of the impurity than that of the pure metal
- d) The low melting point metal is purified by this method

146. NaHSO_4 is reacted with NaCl at 823 K. The liberated gas is reacted with Ca_3P_2 to form CaCl_2 and X. Which one of the following is X?

Options :

- a) PH_3

- b) PCl_5
- c) PCl_3
- d) H_3PO_4

147. The correct order of boiling points of the given hydrides of 16th group elements is

Options :

- a) $H_2O > H_2Te > H_2Se > H_2S$
- b) $H_2Te > H_2Se > H_2S > H_2O$
- c) $H_2Te > H_2Se > H_2O > H_2S$
- d) $H_2O > H_2S > H_2Se > H_2Te$

148. When xenon hexafluoride is completely hydrolysed a xenon compound 'A' is formed. The number of σ -bonds, π - bonds and lone pairs present on Xe in 'A' respectively are

Options :

- a) 4,4,0
- b) 3,3,1
- c) 4,1,1
- d) 4,2,0

149. The spin-only magnetic moment of an ion is 3.87BM. The ion is

Options :

- a) Ti^{2+}
- b) Co^{2+}

- c) Cu^{2+}
- d) Mn^{2+}

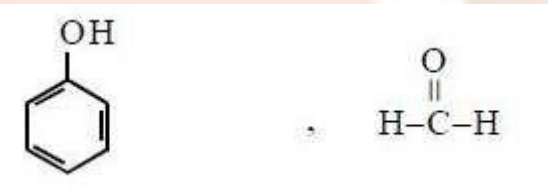
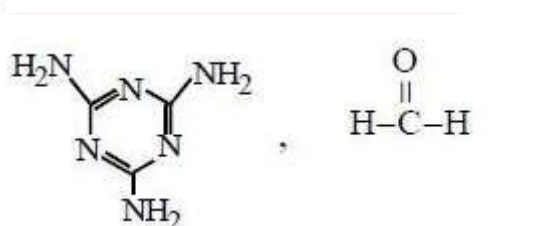
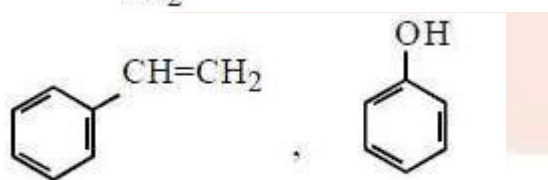
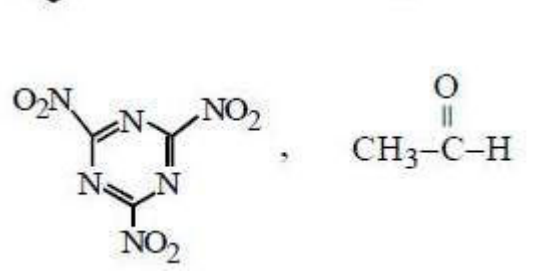
150. Structure of $\text{Cr}_2\text{O}_7^{2-}$ has

Options :

- a) Two tetrahedra sharing one corner
- b) Two tetrahedra sharing one edge
- c) Two tetrahedra sharing one face
- d) Irregular octahedral

151. The monomers required for the formation of melamine polymer are

Options :

- a) 
- b) 
- c) 
- d) 

152. Identify the heterocyclic bases present in DNA from the following

- a) 5 - Methyl -2,4- dioxy pyrimidine
- b) 2,4- dioxy pyrimidine
- c) 2 - oxy - 4 – aminopyrimidine

Options :

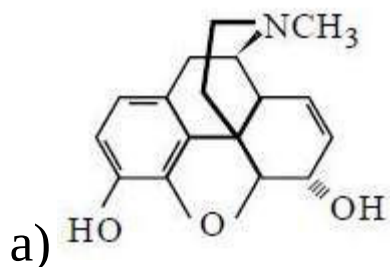
- a) a,b
- b) a,c
- c) b,c
- d) a,b,c

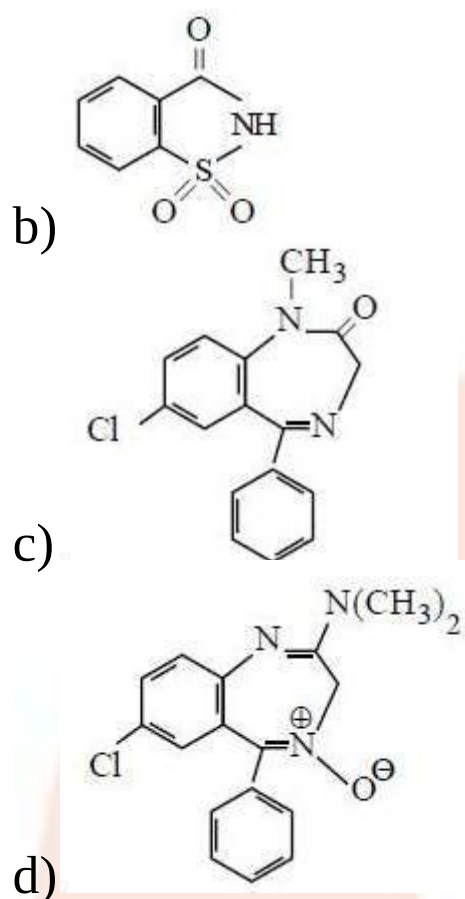
153. Which one of the following is hypnotic?

Options :

- a) Amytal
- b) Ranitidine
- c) Dimetapp
- d) Aspirin

154. Identify an analgesic from the following





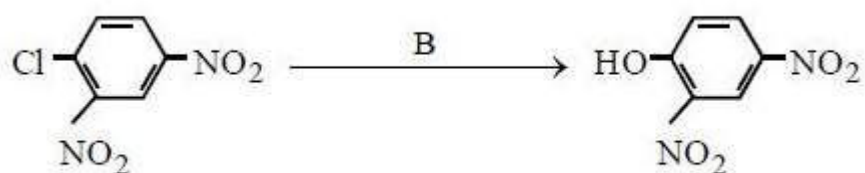
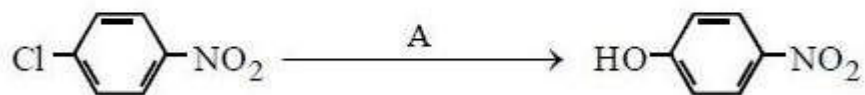
155. Identify the correct statements from the following

- a) Dichloromethane is used as a solvent
- b) Freons are used for refrigeration
- c) CCl_4 causes depletion of ozone layer
- d) Electrophilic substitution reactions of halobenzenes occur faster than those of benzene

Options :

- a) b,c,d
- b) c,d
- c) a,b,c
- d) a,c

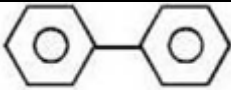
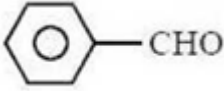
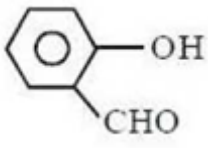
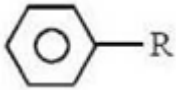
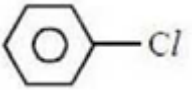
156. identify the reaction conditions for the following reactions



Options :

- | | A | B |
|----|--------------------------|--------------------------|
| a) | i) NaOH/443K
ii) H(+) | i) NaOH/623K
ii) H(+) |
| b) | i) NaOH/443K
ii) H(+) | i) NaOH/368k
ii) H(+) |
| c) | i) NaOH/368k
ii) H(+) | i) NaOH/443K
ii) H(+) |
| d) | i) NaOH/623K
ii) H(+) | i) NaOH/368k
ii) H(+) |

157. Match the following

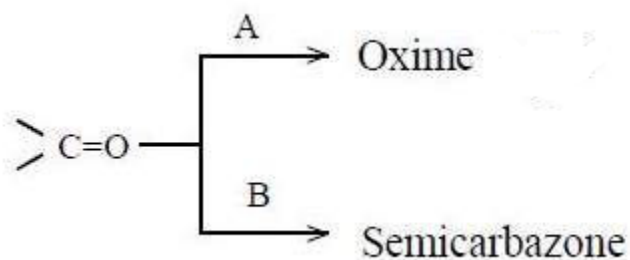
List - I (Name of the reaction)	List - II (Major product)
a) Remiei-Tiemann reaction	i) 
b) Fittig reaction	ii) 
c) Etard reaction	iii) 
d) Sandmeyer reaction	iv) 
	v) 

The correct answer is

Options:

- | | | | | |
|----|-----|----|-----|-----|
| | a | b | c | d |
| a) | iv | ii | iii | i |
| b) | iii | ii | i | v |
| c) | ii | i | iv | iii |
| d) | iii | i | ii | v |

158. In the above reactions A and B respectively



Options:

- a) $\text{H}_2\text{N}-\text{CH}_3$, $\text{H}_2\text{N}-\text{OH}$
- b) $\text{H}_2\text{N}-\text{NH}_2$, $\text{NH}_2-\text{NH}-\text{C}_6\text{H}_5$
- c) $\text{H}_2\text{N}-\text{NH}-\text{CO}-\text{NH}_2$, NH_2-OH
- d) $\text{H}_2\text{N}-\text{OH}$, $\text{H}_2\text{N}-\text{NH}-\text{CONH}_2$

159. The correct order of the groups - CN (a), - Cl (b), - CF_3 (c) and - NO_2 (d) in increasing the strength of a carboxylic acid, when they are bonded to α - carbon of carboxylic acid separately is

Options:

- a) $a < b < c < d$
- b) $d < a < b < c$
- c) $b < a < d < c$
- d) $b < d < a < c$

160. Which of the following statements are correct?

- I. The activating effect of $-\text{NHCOCH}_3$ is more than that of $-\text{NH}_2$ group
- II. N - ethylbenzene sulphonamide is soluble in alkaline solution
- III. Aniline on reaction with concentrated H_2SO_4 followed by heating to 453 - 473 K with concentrated H_2SO_4 gives p-aminobenzene sulphonic acid
- IV. Alkyl isocyanide on reaction with HgO gives alkyl isocyanate

Options:

- a) I, II, IV
- b) III, IV
- c) II, III, IV
- d) II, III

